# AMRIT PUHAN

amrit.puhan@outlook.com • +1 (814) 769-0850 • amritpuhan.com
linkedin.com/in/amritpuhan • github.com/amrit19 • Google Scholar


## SUMMARY


Interested in how reliable collective judgments can be drawn from noisy, disagreeing people, and how that question carries over to evaluating and aligning AI systems with human preferences.


## EDUCATION

**Master of Science, Informatics (Data Science Concentration)** 08/2022 – 08/2024
Pennsylvania State University, University Park, PA, USA GPA: 3.88/4.0
Thesis: *Recovering Ground Truth Rankings When the Majority Is Misinformed* (**Best MS Thesis Award in AI**)
Advisor: Dr. Hadi Hosseini

**Bachelor of Technology, Computer Science and Engineering** 07/2016 – 05/2020
National Institute of Technology, Rourkela, India GPA: 3.7/4.0
Thesis: *Human Gait as a Cognitive State Predictor*
Advisor: Prof. Anup Nandy

## PUBLICATIONS

*First author; authors listed alphabetically. †Equal contribution, co-first author.

1. Hosseini, H., Mandal, D., **Puhan, A.*** SP-Rank: A Dataset for Ranked Preferences with Secondary Information. *KDD 2026*, Datasets and Benchmarks Track (Poster).
2. Hosseini, H., Mandal, D., **Puhan, A.*** Surprisingly Popular Voting with Concentric Rank-Order Models. *The ACM Web Conference (WWW) 2025*, Main Track (Poster), pp. 3026–3036.
3. Hosseini, H., Mandal, D., **Puhan, A.*** The Surprising Effectiveness of SP Voting with Partial Preferences. *NeurIPS 2024*, Main Track (Poster), pp. 2787–2829.
4. Hazra, S., Pisipati, M., **Puhan, A.**, Nandy, A., Scherer, R. Two Novel Methods for Multiple Kinect v2 Sensor Calibration. *CVIP 2021*, Springer, pp. 403–414.
5. Pisipati, M.†, **Puhan, A.**†, Kumar, A., Semwal, V.B., Agrawal, H. A Dynamic Hand Gesture-Based Password Recognition System. *ICCIS 2019*, Springer, pp. 21–34.

## RESEARCH EXPERIENCE

**FAIR Lab, Penn State**, Graduate Researcher 01/2023 – 12/2024
Advised by Dr. Hadi Hosseini; mentored by and collaborated with Dr. Debmalya Mandal (University of Warwick)

*Recovering Ground Truth Rankings When the Majority Is Misinformed (MS Thesis)*

- Designed scalable SP voting algorithms that recover ground-truth rankings from partial preferences when experts are a minority. The best variant (Partial-SP) recovers the true ranking about **4x** more accurately than Copeland on votes alone. *(NeurIPS 2024)*
- Modeled voter behavior as concentric mixtures of Mallows and Plackett-Luce models, inferred parameters with Bayesian MCMC (No-U-Turn Sampler), and derived when SP voting recovers the ground truth with high probability. On real data, SP voting made about **6x** fewer ranking errors than Copeland. *(NeurIPS 2024, WWW 2025)*
- Released **SP-Rank**, the first large-scale public dataset pairing personal votes with predictions of others' votes. It holds **12,384** datapoints from **1,152** participants across 9 elicitation formats, open-sourced with benchmarks showing SP voting recovers rankings about **5x** more accurately than standard Copeland voting. *(KDD 2026)*

*LLM fine-tuning and preference learning*

- Used SP voting to generate preference labels for RLHF reward-model training, with weak and strong language models as voters, improving label accuracy by **9%** on average over standard voting rules in preliminary experiments. *(MS Thesis)*
- Fine-tuned LLMs for multiple-choice QA with LoRA, quantization and PEFT. Ranked **6th** on the AI2 SWAG leaderboard with **90.98%** accuracy (2024).

**Intelligent Computing and Computer Vision Group, NIT Rourkela**, Undergraduate Researcher 08/2019 – 05/2020
Bachelor's thesis advised by Prof. Anup Nandy

- Proposed two calibration methods for an 8-sensor Microsoft Kinect v2 gait-capture setup, reaching a mean reprojection error of **0.42 pixels**, a **76.27%** reduction over existing methods. *(CVIP 2021)*

- Predicted cognitive state from calibrated gait features with ML models ($R^2 = \mathbf{0.9556}$), suggesting gait is a strong indicator of cognitive state.
- Co-developed a hand-gesture password recognition system that runs on an ordinary laptop webcam, reaching **96.16%** test accuracy and outperforming prior webcam (93.75%) and Kinect-based (86.5%) systems. *(ICCIS 2019)*

## TEACHING EXPERIENCE

**Penn State College of IST**, Instructional Assistant, IST 261: Application Development (Java, MVC) Spring 2023

- Coached a 20+ student undergraduate class in object-oriented design, MVC architecture and debugging through office hours and one-on-one assignment help.
- Graded assignments with constructive feedback and served as a liaison between students and faculty.

## ACADEMIC SERVICE

Reviewer, Internal Amazon Science Conference 2026
Reviewer, KDD 2026, Datasets and Benchmarks Track
Reviewer, The ACM Web Conference (WWW) 2025, Main Track

## WORK EXPERIENCE

**Amazon Web Services**, Software Development Engineer II, Applied AI Solutions (Seattle, WA) 12/2024 – Present

- Build AI systems over **petabyte-scale** automotive and industrial data at **sub-second latency**.
- Develop a multi-modal video search system for autonomous-driving (ADAS/AV) teams, helping developers surface rare edge cases across video, sensor and annotation data with plain-English queries.
- Develop time-series anomaly detection for predictive maintenance of industrial equipment, evaluating supervised and unsupervised statistical and deep-learning models.

**Penn State**, Graduate Assistant, Organizational Effectiveness (State College, PA) 08/2023 – 08/2024

- Built Power BI dashboards for tracking student enrollments in Penn State World Campus and a reusable template that cut development time by **60%** and improved tracked enrollment KPIs by **25%**.

**SAP Labs India**, Associate Software Developer (Bengaluru, India) 07/2020 – 07/2022

- Developed **five** front-end applications for SAP SuccessFactors (enterprise B2B cloud).
- Quality Analyst for **six** projects; coordinated the product-wide quarterly testing event **twice**.
- CI/CD contact for **ten** teams; automated Jenkins deployments of **40+** applications across **five** countries.

**Schneider Electric India**, Summer Research Intern (Bengaluru, India) 05/2019 – 07/2019

- Built a cross-platform IoT door-entry prototype (web, Android, iOS) with real-time text, audio and video, cutting latency by **83.3%**.

## HONORS AND AWARDS

- **Best MS Thesis Award in Artificial Intelligence**, third annual Penn State AI Awards (2025).
- **NeurIPS 2024 Scholar Award.** Given to select early-career researchers; grant covering registration and travel.
- **Fast Learner Award**, SAP Labs India (2021).
- **Top 1%, JEE Main.** Among 1.4 million candidates (Ministry of Education, Govt. of India).
- **National Talent Search Examination Scholar**, NCERT (2012). National rank 245 of 1M+.
- **Junior Mathematics Olympiad Scholarship**, IMA, Govt. of Odisha (2012). State rank 8.